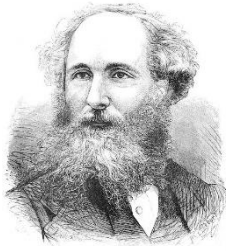


**Nodal Method** continuation...

Steps to determine node voltages:

- 1 Select a node as reference node. Assign voltages to the remaining $n-1$ nodes. These voltages are referenced with respect to the reference node.
- 2 Apply KCL to each of the $n-1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
- 3 Solve the resulting simultaneous equations to obtain the unknown node voltages.

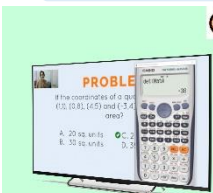
Maxwell's Loop (Mesh) Method or Method of Mesh Current

James Clerk Maxwell

This method was first proposed by Scottish scientist, James Clerk Maxwell using the voltage law only. The method involves a set of independent loop currents assigned to as many meshes as existing in the circuit, and these currents are employed in connection with appropriate resistances when the Kirchhoff's voltage law equations are written. The arbitrarily assumed loop currents may or may not exist in the various resistors but when determined will readily yield the desired values by simple algebraic additions. The scheme offers the advantage that fewer equations need be written to solve a given problem.

Steps to determine mesh currents

- 1 Assign mesh currents to the meshes. A mesh is a loop which does not contain any other loops within it.
- 2 Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
- 3 Solve the resulting n simultaneous equations to get the mesh currents.



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Superposition Theorem

The superposition theorem states that:

"the voltage across or current through an element in a linear circuit is the algebraic sum of the voltages across or currents through that element due to each independent source acting alone".

The idea of the theorem lies on the linearity property.

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The three steps to apply superposition theorem.

- 1 Turn off all independent sources except one source. Find the output (voltage or current) due to the active source using conventional techniques like Ohm's law and voltage and current division theorems.
- 2 Repeat step 1 for each of the other independent sources.
- 3 Find the total contribution by adding algebraically all contributions due to the independent sources.

Thevenin's Theorem

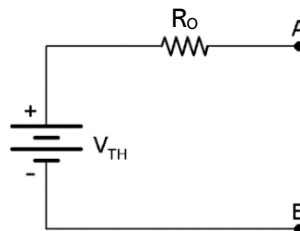
This theorem was developed by M. Leon Thevenin in 1883, a French Telegraph engineer.



The Thevenin's theorem states that:

"Any network with two open terminals a and b can be replaced by a single voltage source V_{TH} in series with a single resistance, R_o ".

Below is the Thevenin's equivalent circuit which is a simple series circuit.



where:

 V_{TH} or V_{OC} = Thevenin's voltage or open-circuit voltage R_o = looking back resistance

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Norton's Theorem

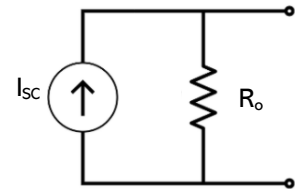
This theorem is basically similar to the Thevenin's theorem. This was proposed by Edward Lawry Norton in 1926, an American engineer at Bell Telephone Lab.



The Norton's theorem states that:

"Any network with 2 open terminals a and b can be replaced by a single current source, I_{SC} in parallel with a single resistance, R_o ".

The Norton's equivalent circuit, a parallel circuit, is shown below.



where:

 I_{SC} = short-circuit current R_o = looking back resistance**Source Transformation Method (Thevenin-to-Norton conversion)**

The Thevenin's equivalent circuit (a voltage source) can be transformed to a Norton's equivalent circuit (a current source) using the formulas below:

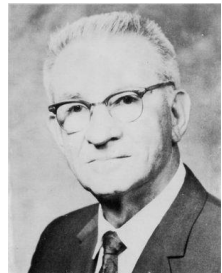
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$$R_{Norton} = R_{Thevenin} = R_o$$

$$I_{SC} = \frac{V_{Th}}{R_{Th}}; V_{Th} = I_{SC} R_{Th}$$

Millman's Theorem

Also known as **Parallel Generator Theorem**, this is used to compute the voltage at the ends of a circuit made up of only branches in parallel. It is named after Jacob Millman, who proved the theorem.



"When any number of voltage sources of arbitrarily generated voltage and finite internal resistance different from zero are connected in parallel, the resulting voltage across the parallel combination is the ratio of the algebraic sum of the currents that each source individually delivers when short-circuited to the algebraic sum of the internal conductance."